Fathima Rizmiya

AI / Machine Learning Engineer

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SUMMARY

MSc graduate in Artificial Intelligence with practical experience gained through projects in machine learning, deep learning, and LLM-powered applications. Enjoys solving problems with data and building useful AI applications, with additional experience in Python, React, and Next.js development. Brings a strong scientific background, analytical thinking, and the ability to explain complex ideas clearly, developed through both technical projects and seven years of science education.

TECHNICAL SKILLS

- **Programming:** Python, JavaScript, TypeScript
- **AI/ML:** PyTorch, PyTorch Geometric, TensorFlow, Keras, scikit-learn, graph neural networks, explainable AI, NLP, feature engineering, hyperparameter tuning
- **LLMs and data:** LangChain, OpenAI API, Gemini API, DeepSeek API, NumPy, Pandas, RDKit, SARIMA, Power BI
- **Development and Deployment:** FastAPI, React, Next.js, Tailwind CSS, MongoDB, Prisma, Git, GitHub

PROJECTS

Interpretable GNN for Toxicity Prediction | *PyTorch, PyTorch Geometric, RDKit* | GitHub — MSc thesis

- Developed a motif-aware GINE graph neural network to predict molecular toxicity on Tox21, achieving a ROC-AUC of 0.817 using a scaffold-based data split.
- Applied Integrated Gradients to interpret predictions and examined explanations against SMARTS based chemical alerts.

AI Research Agent | *Python, LangChain, Gemini, FastAPI* | Live demo | GitHub

- Developed a tool-using LLM agent that selects web search or Wikipedia tools to generate concise, source supported research summaries.
- Exposed the agent through a FastAPI web interface and command line interface.

Climate Forecasting and Power BI | *Python, Pandas, SARIMA, Power BI* | GitHub

- Developed SARIMA forecasting models using ten years of NOAA temperature data from six Sri Lankan stations, achieving a national RMSE of 0.83 °C.
- Presented forecasting results through interactive Power BI dashboards.

YouTube Comment Sentiment Analyzer | *TensorFlow, Keras* | GitHub

- Trained a Keras model that classifies comments fetched through the YouTube Data API as positive, negative or neutral, shown in a Flask web app.

Full-Stack Web Apps | *Next.js, TypeScript, MongoDB, Prisma, NextAuth, Stripe*

- Developed PostHub, a blogging platform with authentication and user dashboards, and ElectroSwift, an e-commerce application with Stripe checkout and an administrative dashboard.

EXPERIENCE

Web Developer, AI Integration | *Bookolog.lk, Sri Lanka (remote)* — 2023 – Present

- Integrated a DeepSeek-powered AI shopping assistant to support book discovery and answer customer queries about delivery, returns, and payments.
- Contributed to developing and maintaining e-commerce features using the platform's web technologies.

Science Educator | *Universities and secondary schools, Sri Lanka* — 2016 – 2022

- Taught science across university and secondary-school settings, including laboratory instruction, lectures, and student mentoring.
- Developed strong communication and analytical skills through explaining complex scientific concepts to diverse audiences.

EDUCATION

MSc in Artificial Intelligence (Hons) | *Ural Federal University, Russia* — 2024 – 2026

- Thesis: an interpretable, motif-aware graph neural network for molecular toxicity prediction.

Bachelor of Science (Hons) | *University of Peradeniya, Sri Lanka* — 2011 – 2015

- Four year honours degree, majoring in Chemistry with a minor in Computer Science.

CERTIFICATIONS

- **Machine Learning Specialization**, DeepLearning.AI and Stanford University (Coursera), 2026
- **React Development**, General Assembly, Melbourne, 2023
- All credentials: linkedin.com/in/fathima-rizmiya/details/certifications

Languages: English, Tamil, Sinhala